

Neural Network Option Pricing

Approximating Black-Scholes European Call Prices with Feed-Forward Neural Networks

Objective

Evaluate whether a feed-forward neural network can learn the Black-Scholes European call pricing function and act as an efficient pricing surrogate.

Project group

Machine Learning for Finance

Research Question

Research question

Can a feed-forward neural network accurately approximate the Black-Scholes pricing function for European call options?

The project compares:

- ▶ the analytical Black-Scholes formula;
- ▶ Monte Carlo simulation;
- ▶ a supervised neural network pricing surrogate.

Black-Scholes SDE

The underlying asset follows the geometric Brownian motion

$$dS_t = rS_t dt + \sigma S_t dW_t,$$

where r is the risk-free rate, σ is volatility and W_t is a Brownian motion.

The analytical European call price is used as the ground truth:

$$S_T = S_0 \exp \left(\left(r - \frac{1}{2} \sigma^2 \right) T + \sigma W_T \right).$$

European Call Option Pricing

A European call option has payoff

$$\Phi(S_T) = \max(S_T - K, 0).$$

Under the Black-Scholes model, its analytical price is

$$C_{BS} = S_0 N(d_1) - Ke^{-rT} N(d_2),$$

with

$$d_1 = \frac{\log(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}.$$

Synthetic Dataset

Synthetic observations are sampled over the input space:

$$x = (S_0, K, T, r, \sigma), \quad y = C_{\text{BS}}(S_0, K, T, r, \sigma).$$

Variable	Range
Initial price S_0	[50, 150]
Strike K	[50, 150]
Maturity T	[0.1, 2]
Risk-free rate r	[0, 0.05]
Volatility σ	[0.1, 0.6]

Neural Network as Pricing Surrogate

The neural network learns the pricing map

$$f_{\theta} : (S_0, K, T, r, \sigma) \mapsto C_{BS}.$$

Baseline architecture:

- ▶ fully connected feed-forward network;
- ▶ ReLU hidden activations;
- ▶ mean squared error loss;
- ▶ Adam optimizer;
- ▶ train/validation/test split with feature scaling.

Experimental Pipeline

1. Sample synthetic contracts.
2. Compute Black-Scholes analytical prices.
3. Train the neural network on the supervised pricing task.
4. Estimate selected prices with Monte Carlo simulation.
5. Compare errors and analyse where approximation is harder.

Evaluation Metrics

The main quantitative metrics are:

- ▶ Mean Absolute Error;
- ▶ Root Mean Squared Error;
- ▶ coefficient of determination R^2 ;
- ▶ relative error diagnostics where meaningful.

Error analysis is performed against moneyness S_0/K , maturity T , and volatility σ .

Main Results

To be completed after the final experiment

This slide will report the main test-set metrics for the neural network and the numerical benchmark.

Method	MAE	RMSE	R^2
Neural network	–	–	–
Monte Carlo	–	–	–

Error Analysis

Figure not yet available:
`../outputs/figures/training_history.pdf`

Monte Carlo Benchmark

Simulated terminal prices are generated as

$$S_T = S_0 \exp \left(\left(r - \frac{1}{2} \sigma^2 \right) T + \sigma \sqrt{T} Z \right), \quad Z \sim \mathcal{N}(0, 1).$$

The discounted estimator is

$$\hat{C}_{\text{MC}} = e^{-rT} \frac{1}{M} \sum_{m=1}^M \max(S_T^{(m)} - K, 0).$$

Discussion

Expected discussion points:

- ▶ approximation quality of the neural network surrogate;
- ▶ regions of the input space with larger errors;
- ▶ comparison with Monte Carlo as a numerical benchmark;
- ▶ computational trade-offs between analytical, simulation-based and learned pricing.

Limitations

- ▶ The first version uses synthetic data generated under Black-Scholes assumptions.
- ▶ The model prices European calls only.
- ▶ The target function is known analytically, so the neural network is evaluated as a surrogate rather than as a replacement for financial theory.
- ▶ Generalization outside the sampled parameter ranges is not guaranteed.

Conclusions and Future Work

The project evaluates neural networks as pricing surrogates in a controlled financial setting.

The central question is not whether the neural network replaces Black-Scholes, but whether it can learn its pricing function accurately and efficiently from supervised data.

Future extensions may include stochastic volatility, Heston dynamics, American options and Greeks estimation.